

Careers In Information Technology

Rosemary Y. Gallardo-Sifuentes and Muhammad Mahad

Northern Virginia Community College

ITE152: Introduction to Digital and Information Literacy and Computer Applications

Nasser A. Maksoud

12 October 2026

Careers In Information Technology

Careers in Information Technology are vastly growing, especially in this age of technology. One way to stay on top of all this new technology is to be a part of the brilliant minds behind it in a job like: Computer and Information Research Scientist. These scientists work at a theoretical level to design algorithms that develop computer language, create programs, and help simplify the way people interact with computers. However, as more computers and technology are made, there has been an increase in cyberattacks. A good way to keep myself and others safe would be working as an Information Security Analysts.

Computer and Information Research Scientists

These scientists work in the research field to improve and make new technology. They explore and address issues within the computational world¹. They design, conduct, and analyze experiments and publish papers on their research. There are certain specialties such as programming and robotics. Programming focuses on the language of computers while robotics focuses on the physical machine. There is a 22% outlook for an increase in this field, which is very good. With an average annual wage of \$140,300 (U.S. BUREAU OF LABOR STATISTICS, 2018). This would definitely be a very secure job. However, there are some cons to consider such as more schooling; at minimum this job requires a master's degree, but most have a PhD. Which means devoting time and money. While it is possible to find a job with a bachelor's in computer or technology related, those are mainly government jobs that don't pay as well. Moreover, a big part of any field in science is collaboration, which depending on the person could be a pro or a con. Personally, I find it a bit hard to work on projects with people if not

¹ Refers to a society that heavily relies on and uses computers, algorithms, data, and digital systems.

everyone is on board, but I also enjoy working with others because it breeds more creative ideas. While more education can turn someone away from this field it should be considered that more education will open up more opportunities and could bring a higher income. Some employers will even pay for the degree which is a win for both parties. This is a strong career choice for any scientist interested in technology.

A Typical Day Working as a Computer and Information Research Scientist

Many computer scientists start off their morning by checking their emails and responding to emails. By midday they focus on designing, setting up, or conducting their experiment. Which is then followed by gathering and analyzing data. A big part of their day; whether it is during, before, or after the experiments; are talking and collaborating with their colleagues. Days can be different depending on the mission. Some days are simply identifying problems, finding solutions, attending meetings, or writing their research paper. One thing is for certain: the job continues even at home. These scientists must stay up to date with current and prospective technology advancements (OutstandingStar, 2024).

Information Security Analysts

Information security analysts are in charge of protecting organizations' computer systems and networks from online threats. They plan and execute security measures, monitor networks for security breaches, investigate breaches, and handle the maintenance of systems. Additionally, they use security tools and software like firewalls and data-encryption programs², create security standards, write reports on breaches, and assist users in understanding procedures (U.S. BUREAU OF LABOR STATISTICS, 2024). Most information security analyst roles require a bachelor's degree in computer and information technology or related fields. Additional

² A software which converts files or messages into a secret code that is only readable by people with the proper decryption key.

information security certifications are recommended by employers and strengthen candidate profiles. Overall, this career holds a 21% projected employment growth between the years 2025 to 2035 and its associated median annual wage in the state of Virginia stands at \$134,900 (U.S. BUREAU OF LABOR STATISTICS, 2024).

Aside from its high projection of employment growth within the next decade and median annual wage, there are various benefits associated with being an information security analyst. Some examples include the ability to specialize in specific fields and opportunity to advance leadership positions such as a chief security officer or computer and information systems manager. Most information security analysts work 40 hours a week, while some may work over 40 hours (U.S. BUREAU OF LABOR STATISTICS, 2024). One con related to this career is that call-ins during unscheduled hours may be necessary in case of security emergencies. Another difficulty can be the need to stay up to date with technology and methods used by attackers. Ultimately, an information security analyst is a well-perceived career.

A Typical Day Working as an Information Security Analyst

A typical workday in the life of an information security analyst consists of monitoring organizational computer networks for security breaches and viruses and conducting investigations into security issues. Analysts perform system checks, risk assessments, and protect files from unauthorized access. A key task also includes creating documentation which can be used to develop procedures, write reports, and train users about rules and processes. Communication is an important everyday skill used by IS analysts because they must share information with supervisors, co-workers, and others about their findings or issues (ONET OnLine, 2024). Typically, information security analysts are always learning and improving their knowledge and abilities.

Conclusion

Both information security analysts and information research scientists are careers which are filled with opportunities in the field of information technology. While information research scientists focus on creating and improving technology with research, information security analysts protect that technology for businesses and general users. After researching both careers, the career that works the best for both of us (Muhammad Mahad and Rosemary G. Sifuentes) is an information security analyst. This career works for us because it provides a constant learning experience, a further opportunity for growth, and combines security with problem-solving skills.

References

Information security analysts. (2021). Wwww.Onetonline.Org.

<https://www.onetonline.org/link/summary/15-1212.00>

This source gives an insight into different areas of being an information security analyst such as requirements, work activities, and opportunities.

CareerOneStop. (n.d.). Wwww.Careeronestop.Org. Retrieved October 7, 2026, from

<https://www.careeronestop.org/videos/careeronestop-videos.aspx?videocode=15121200>

This source was used to find data on the estimated salary (annual wage) for an 'information security analyst' in the state of Virginia.

OutstandingStar. (2024, June 12). *A day in the life of a Computer & Information Research*

Scientist - OutstandingStar.Com. OutstandingStar.Com. <https://outstandingstar.com/a-day-in-the-life-of-a-computer-information-research-scientist/>

This was referenced to explore the day in the life of an employee working in computer and information research.

U.S. BUREAU OF LABOR STATISTICS. (2018, April 13). *Computer and Information*

Research Scientists. U.S. Bureau of Labor Statistics. Bls.Gov.

<https://www.bls.gov/ooh/computer-and-information-technology/computer-and-information-research-scientists.htm>

This source was referenced for the statistics such as average pay, education, and job outlook for a career in computer and information research.

U.S. BUREAU OF LABOR STATISTICS. (2024, August 29). *Information security analysts: Occupational outlook handbook: U.S. Bureau of Labor Statistics*. U.S. BUREAU OF LABOR STATISTICS. Bls.Gov. <https://www.bls.gov/ooh/computer-and-information-technology/information-security-analysts.htm>

This government source provides real data and statistics within the United States. This source was used to provide various information (employment growth stats, responsibilities, skills, etc.) regarding the career of an information security analyst.